

Piezoelectric micro-jet devices: A review

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ABSTRACT

Piezoelectric micro-jet devices (PMJDs), also known as piezoelectric inkjet printing, are used to distribute the functional materials to form single or complex microstructures on a substrate. As one of the most promising technologies in the field of additive manufacturing in the 21st century, the PMJDs have received a lot of attentions from universities, enterprises and research institutes. In this review, the origin of the PMJDs is introduced and the data analyses of the PMJDs are also counted based on Web of Science database. Concretely, the fundamental actuation modes and structure configurations of the PMJDs are summarized firstly. Secondly, the analysis methods of the PMJDs are discussed. Thirdly, the challenges of the PMJDs are explicitly discussed. Fourthly, aiming at the current challenges, the schemes on improving the droplet uniformity and productivity of the PMJDs are discussed in detail. However, there are still many studies to be completed and optimized to solve the bottlenecks faced by the PMJDs. Finally, the applications of the PMJDs are described systematically. In general, this review is expected to be of interest to both seasoned researchers and practitioners in the PMJDs and those entering this field for the first time.

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1. Introduction

1.1. Origin of PMJDs

Modern microstructure products have been rapidly developing towards the directions of more compact and multifunctional, convenient with a higher expectancy of quality and reliability. There are increasing demands for reducing material content and energy consumption during the fabrication process and presenting less pollution to meet the sustainable development [1–4]. Based on this, adding manufacturing technology comes into being due to its high precision, high utilization rate, short development cycle and the ability to manufacture arbitrary complex parts. The additive manufacturing technology, including laser-induced forward transfer [5–7], inkjet printing [8–11], electrostatic spraying [12] and rapid prototyping [13], has been adopted to manufacture microstructure products. Among these methods, inkjet printing is an essential subject for industry and academic research, and the generated droplet can be controlled to an accuracy of picoliter level and can also be placed onto a substrate to an accuracy of micrometer. Inkjet printing has emerged as one of the most powerful tools for graphics and printing operations [14–18] and has been gradually developing for displays, MEMS, cell printing, biochips, LCD fabrication, ceramic components, tissue engineering, solar cell and nanotechnology [19–24].

Inkjet printing includes continuous jet (CJ) and drop-on-demand jet (DODJ). The CJ based on the transducer, charging electrode and electric field can produce the droplet continuously, and the droplet location on a substrate can be determined by its charging density. A catcher device needs to be provided to catch the uncharged droplets. Instead of the CJ, it is possible to produce a DODJ droplet only utilizing an actuation pulse. The advantage of the DODJ over the CJ is that there is no need for the complicated hardware for the break-off synchronization, charging and deflection electrode, guttering, recirculation system, high-pressure fluid supply and complicated circuitry. Table 1 shows the performance

Table 1
Comprehensive performance comparisons between the CJ and DODJ.

Jet type	Velocity	Accuracy	Cost	Availability	Operation	Compatibility
CJ	Fast	Low	High	Low	Complex	Poor
DODJ	Slow	High	Low	High	Simple	Good

Table 2
Comprehensive performance comparisons between the TJDs and PMJDs.

Jet type	Velocity	Accuracy	Cost	Stability	Lifetime	Jetted object	Structure
TJDs	Slow	Low	Low	Low	Short	Less	Simple
PMJDs	Fast	High	High	High	Long	More	Complex

comparisons between the CJ and DODJ [25–28]. Although the CJ can achieve high velocity, the DODJ has the advantages of simple operation, low cost and high accuracy.

There are several kinds of actuators for the DODJ device, including piezoelectric, thermal, solenoid, pneumatic, magnetostrictive and acoustic actuators [29–31]. Among these actuators, thermal jet devices (TJDs) and piezoelectric micro-jet devices (PMJDs) are the most common ways to distribute the functional materials because of high-frequency response, controllable droplet size and potential capacity to eliminate the satellite droplets. Table 2 shows the comprehensive performance comparisons between the TJDs and PMJDs.

The TJDs are characterized by a larger deformation, simpler design and lower cost, and the droplet is jetted easily from the nozzle by inducing the formation of a vapor bubble in the fluid using resistive film. However, the fluid has to be stable thermally due to the relative dependency drawback between temperature and liquid properties. Generally, the TJDs are only used in the traditional graphics printing. In recent decades, the PMJDs have been rapidly developed because of the prominent advantages of piezoelectric principle [32–40], such as rapid dynamic response, convenient control, anti-electromagnetic interference and simple structure. Compared with the TJDs, the PMJDs need not to create the fluid thermal stress, which increases the lifetime of both the device and the fluid. The droplets can be jetted by applying a voltage pulse to a piezoelectric actuator and the droplet position, velocity and volume can be easily controlled by a function generator. The PMJDs have been widely applied to biomedicine, microelectronic system, material chemistry, electronic packaging, 3D printing and space mechanism lubrication. Fig. 1 shows the origin of the PMJDs in additive manufacturing field.

1.2. Data analyses of PMJDs

Research efforts on the PMJDs have been carried out in the past few decades. We run queries using the common keywords that

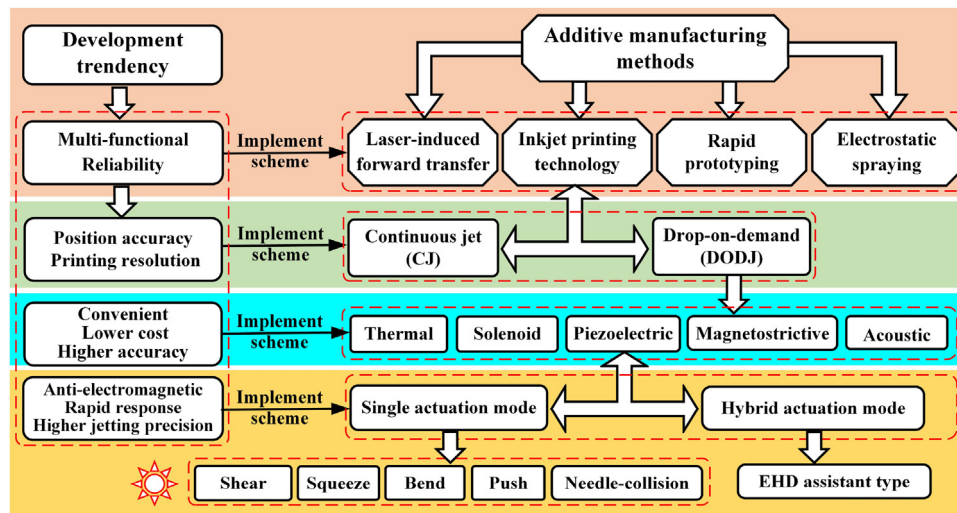


Fig. 1. The origin process of the PMJDs in additive manufacturing field.

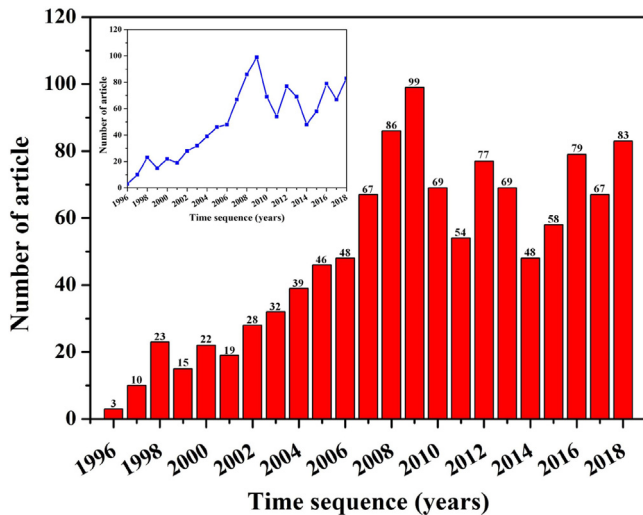


Fig. 2. Annual variation trends of the PMJDs and the number of articles.

related to the research on the PMJDs in Web of Science database, with AND operator(s) between keywords. The keywords include piezoelectric (AND) inkjet, piezoelectric (AND) jet, piezoelectric (AND) injection, piezoelectric (AND) ejection, piezoelectric (AND) dispenser. The retrieval results are obtained on December 31 st, 2018. Fig. 2 shows the annual variation trends of the PMJDs and the number of articles. The development of the PMJDs entered a golden period in 2009. But there has been no breakthrough in recent years. Especially in 2014, the study of the PMJDs entered a bottleneck period. We only counted the number of articles published between 1996 and 2018 because few articles were published before 1996. In fact, the basis of the PMJDs was attributed to three patents in the 1970s [104]. Firstly, a squeeze mode droplet generator was proposed by Zoltan of Clevite company in 1972 (US Patent 3683212). Secondly, a bend type jetting device was designed by Stemme of Chalmers University in 1973 (US Patent 3747120). Thirdly, a bend mode dispensing system was also reported by Kyser and Sears of Silonics company in 1976 (US Patent 3946398).

The relevant researches have been conducted by research institutes and universities in 38 countries across five continents, as shown in Fig. 3, which suggests the global impact of the PMJDs. The country distribution on different continents has been characterized by the red flag. The global research hot on the PMJDs from high to

low are Europe, Asia, North America, Oceania and Africa. Among them, Germany and the UK are the leaders in Europe, while China, Korea and Japan are the leaders in Asia. The USA is the leader in North America. However, a small number of researches have been done in Oceania and Africa.

In this review, we discuss the state-of-the-art of the PMJDs in detail. Firstly, the fundamental actuation modes and structure configurations of the PMJDs are explained. Secondly, the analysis methods of the PMJDs are summarized. Thirdly, the challenges of the PMJDs are discussed. Fourthly, the schemes on improving the droplet uniformity and productivity are highlighted aiming at the bottlenecks. Finally, the applications of the PMJDs are discussed. Moreover, the future developments of the PMJDs are predicted.

2. Actuation modes and configurations of PMJDs

There are two actuation modes for the PMJDs, including single actuation mode and hybrid actuation mode. The single actuation mode includes shear mode, squeeze mode, bend mode, push mode and needle-collision mode, while the hybrid actuation mode refers to electrohydrodynamic (EHD) assistant actuation.

2.1. Single-actuation mode

2.1.1. Shear type

Fig. 4(a) shows the schematic of the shear type PMJD. The d_{15} mode is used to drive the fluid channels. The two adjacent piezoelectric elements bend in opposite directions, and an emerging pressure wave pushes the fluid to the nozzles and jets the droplets. Antohe et al. designed a shear type PMJD and studied the propagation characteristics of pressure wave [41]. Brunahl et al. proposed a shear type PMJD that was composed of a base plate made of polarized PZT and an inactive cover plate made of depolarized PZT, as seen in Fig. 4(b), in which the cover plate was glued onto base plate, and the nozzle plate was assembled onto front surface of base plate [42,43]. Cheng et al. presented a shear type PMJD based on COMS and MEMS post process, in which the pressure chamber immersed the microfluidic channel and the nozzle plate was fabricated on single chip [44].

2.1.2. Squeeze type

Fig. 5(a) shows the schematic of the squeeze type PMJD. The d_{33} mode is used to drive the fluid channel. A small amount of liquid is expelled from the nozzle due to the sudden decrease of the enclosed

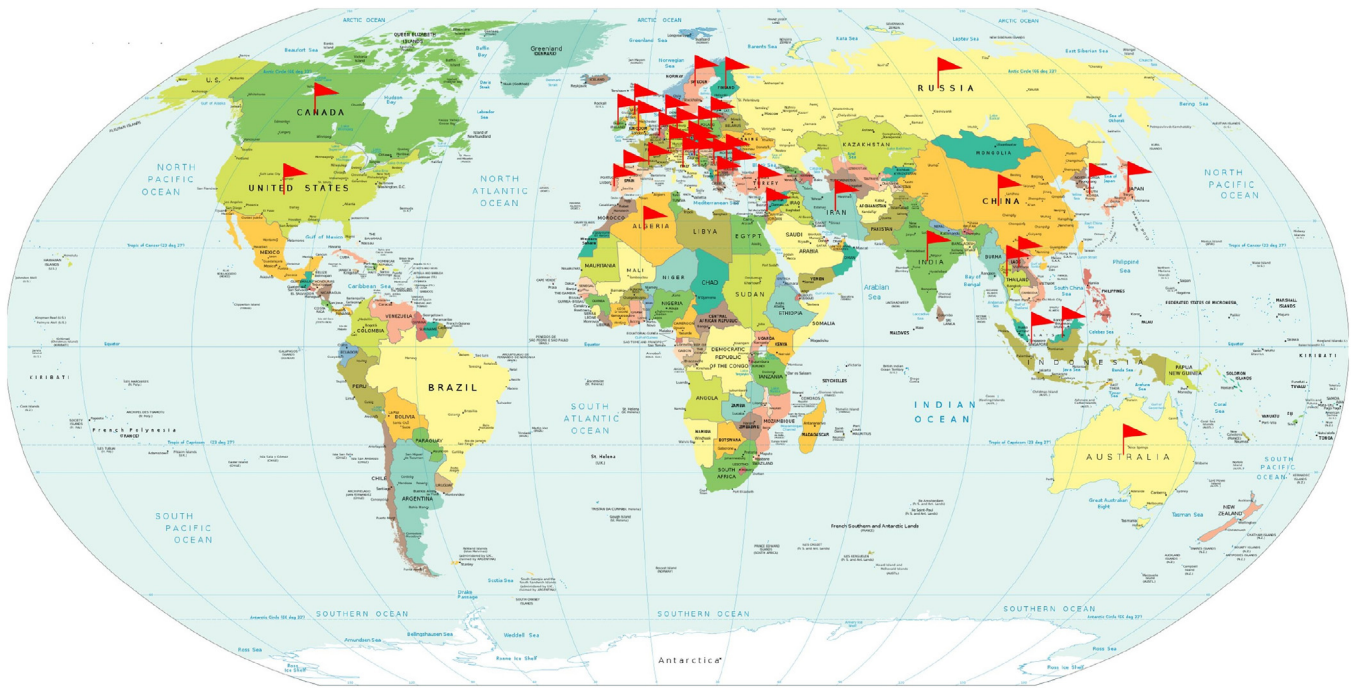


Fig. 3. Conducted researches on the PMJDs by different countries.

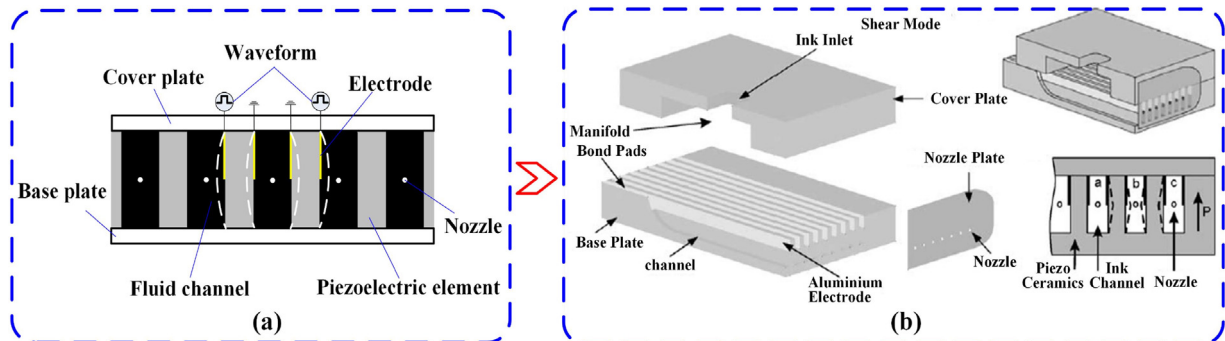


Fig. 4. Actuation mode and configuration of the shear type PMJD: (a) schematic of the shear type PMJD; (b) shear type PMJD by Brunahl et al. [42].

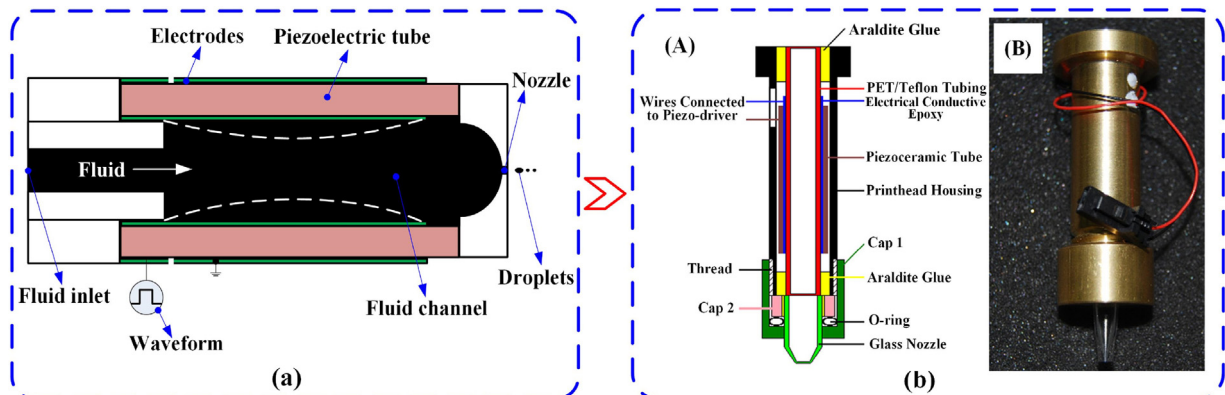


Fig. 5. Actuation mode and configuration of the squeeze type PMJD: (a) schematic of the squeeze type PMJD; (b) squeeze type PMJD by Li et al. [52].

volume. Then, the tube expands slowly to the initial volume and the accompanying pressure reduction is too small to overcome the surface tension at the nozzle. The liquid flows into the tube from the fluid inlet, which replaces the liquid expelled from the nozzle previously. Some researches on the squeeze type PMJDs have been

done [45–54]. Bugdayci et al. presented a squeeze PMJD based on the axisymmetric motion of piezoelectric cylinder [45]. Kung et al. proposed a squeeze type PMJD based on the thin-walled glass tube and cylindrical piezoelectric tube, which could jet stable droplets in the diameter range of $1\ \mu\text{m}$ – $2\ \mu\text{m}$ [48]. A squeeze type PMJD was

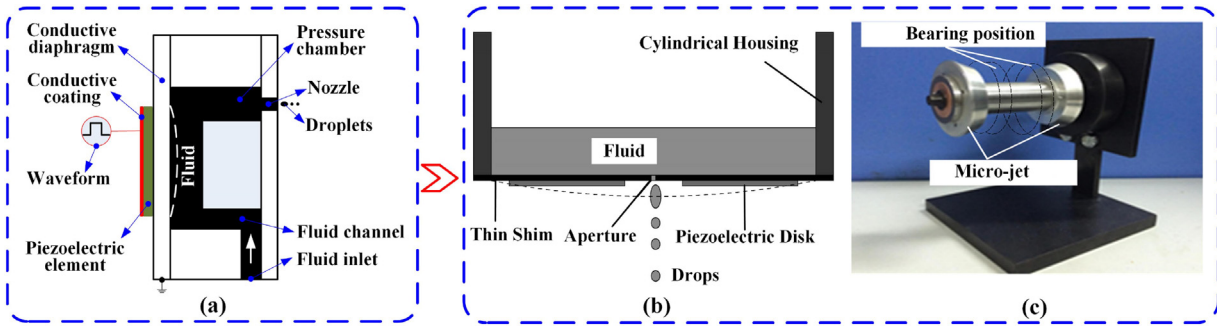


Fig. 6. Actuation mode and configuration of the bend type PMJD: (a) schematic of the bend type PMJD; (b) bend type PMJD by Lam et al. [60]; (c) bend type PMJDs by Liu et al. [67].

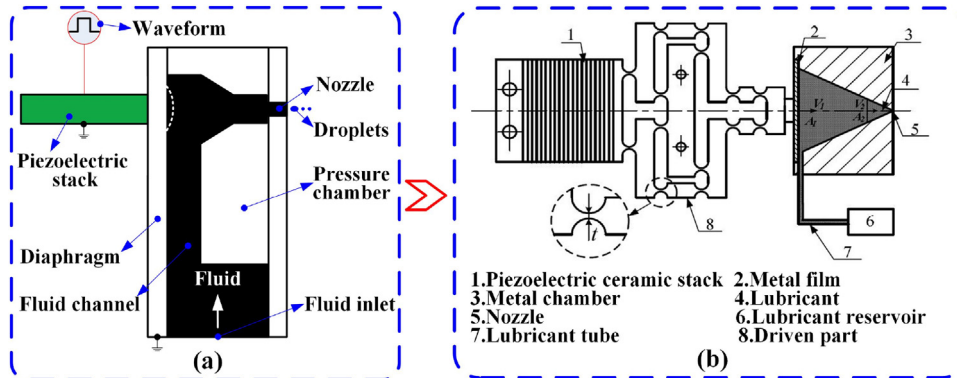


Fig. 7. Actuation mode and configuration of the push type PMJD: (a) schematic of the push type PMJD; (b) push type PMJD by Liu et al. [75].

proposed by Ulmke et al., which generated the droplet with the diameter range $10\ \mu\text{m}$ – $50\ \mu\text{m}$ [50]. Li et al. developed a squeeze type PMJD to distribute the multiple materials on demand, and a synchronizer based on microcontroller was designed to synchronize printing process from the multi-nozzles concerning movable positioning stage [51]. A squeeze type PMJD based on PET/PTFE with interchangeable nozzles was designed and fabricated by Li et al., as seen in Fig. 5(b), which distributed the liquid with the viscosity up to 100 cps [52]. Shin et al. presented a squeeze PMJD that consisted of a control element and a nozzle, and a hidden glass capillary tube with a diameter of $50\ \mu\text{m}$ was wrapped by piezoelectric actuator [53].

2.1.3. Bend type

Fig. 6(a) shows the schematic of the bend type PMJD. The d_{31} mode is used to deform the pressure chamber. The conductive diaphragm bends inwardly into the pressure chamber, and the droplet is jetted from the nozzle. Lots of researches on the bend type PMJDs have been done [55–69]. Yang et al. designed a bend type PMJD which achieved the droplet velocity range from 0.5 m/s to 0.7 m/s [55]. Perçin et al. proposed two bend types of PMJDs to jet water, ink, powder and photoresist, and the droplet volume ranged from 34 fL to 1.5 nL [56,57]. Lam et al. fabricated a bend type PMJD using a PMN–PT single crystal ring [60], as seen in Fig. 6(b). Fan et al. designed a PMJD that could be used for liquid dispensing system [61]. Shen et al. presented a novel bend PMJD based on a ring type piezoelectric ceramic and a cymbal shaped nozzle plate, which could achieve the droplet diameter of $4.07\ \mu\text{m}$ and atomization rate of 0.5 mL/min [62]. Yamaguchi et al. proposed an elliptical bend PMJD with a transparent nozzle tip [63]. Wu et al. designed a PMJD with the interchangeable nozzle, and a clogged nozzle could be removed to clean or replace easily [64]. Sadeghian et al. designed a bend type PMJD, and the droplet behavior was

studied [66]. Liu et al. proposed a novel bearing lubricating device based on the bend PMJDs, as seen in Fig. 6(c), which was embedded in gyroscope's bearing system [67,68].

2.1.4. Push type

Fig. 7(a) shows the schematic of the push type PMJD. The d_{33} mode is also used to deform the pressure chamber. The droplet can be jetted from the nozzle. Some researches on the push type PMJDs have been done [70–76]. Chen et al. proposed a push type PMJD and studied the droplet formation by changing channel pressure [70]. Fuller et al. proposed a push type PMJD for MEMS and IC by printing the metal colloids [71]. Dong et al. presented a push type PMJD with a nozzle diameter of $53\ \mu\text{m}$ [72]. Tzvi et al. designed a push type PMJD that could generate the droplet with high accuracy up to several picoliters [73]. Liu et al. proposed a push type PMJD based on flexible hinges for space mechanism lubrication [75], as seen in Fig. 7(b). Gross et al. proposed a push type PMJD using the pyrex-glass, silicon membrane and piezoelectric stack [76].

2.1.5. Needle-collision type

Fig. 8(a) shows the needle-collision type PMJD. At the initial time, the distributed fluid is filled with the fluid channel, and the needle seals the nozzle. Then, the actuator is applied at a high level to drive the needle move upward. The nozzle is opened and the liquid is jetted under the pressure of reservoir. When the waveform changes from the high level to low level, the needle is driven to move back quickly and cuts off the liquid flow. Relevant researches have been done [77–93]. A new needle-collision type PMJD was presented by Nguyen et al., which could provide a small droplet from 50 cps to 500 cps [77,78]. Jeon et al. proposed a novel PMJD based on two piezo stacks and a lever-hinge, which was applied to light emitting diode and flip chip packaging [80]. Wang et al. proposed a needle-collision PMJD driven by piezo stack, and the

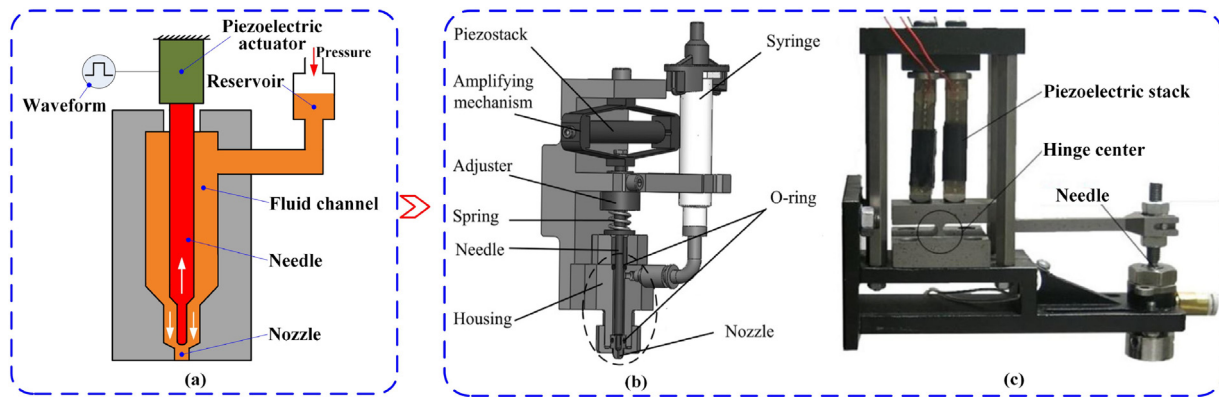


Fig. 8. Actuation mode and configuration of the needle-collision type PMJD: (a) schematic of the needle-collision type PMJD; (b) needle-collision type PMJD by Lu et al. [84]; (c) needle-collision type PMJD by Deng et al. [92].

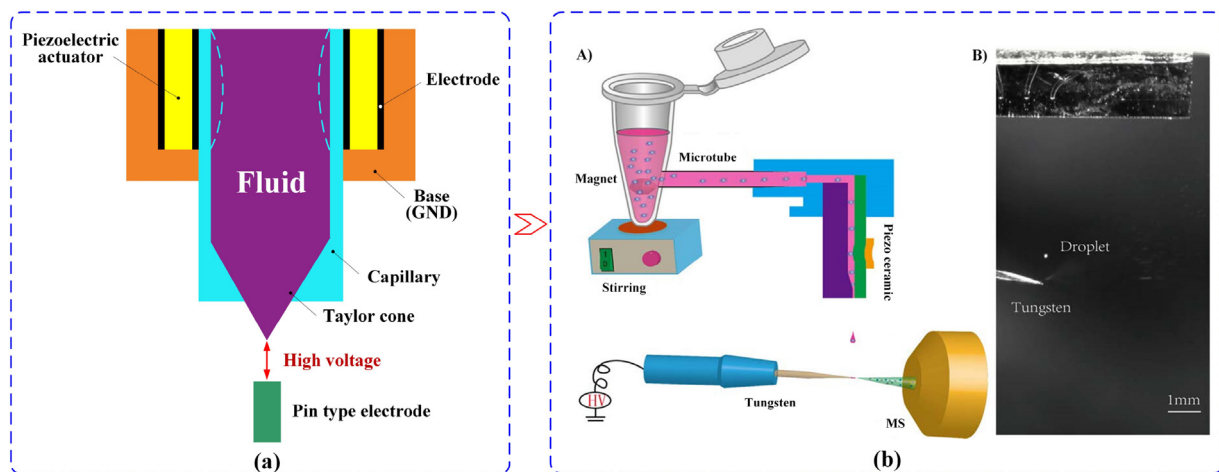


Fig. 9. Actuation mode and configuration of the EHD assisted hybrid actuation type PMJD: (a) schematic of the needle-collision type PMJD; (b) EHD assistant type PMJD by Chen et al. [97].

maximum jetting frequency of 65 Hz and the droplet diameter of 1.07 mm were realized using the stainless steel nozzle with a diameter of 0.25 mm [82]. Besides, the displacement amplifying mechanism was redesigned and the droplet diameter of 525 μm was achieved [83]. Lu et al. proposed a needle-collision PMJD using symmetric rhombus flexure hinge [84,85], as shown in Fig. 8(b), and the fluid viscosity of 58 Pa.s and droplet volume of 0.62 μL were achieved. Zhou et al. proposed a novel PMJD based on bi-piezoelectric actuator, and the maximum jetting frequency were 500 Hz [86]. Besides, the simulation model of the PMJD were also established [89]. Deng et al. proposed a novel PMJD with a pin joint structure, as seen in Fig. 8(c), and the adhesives with a viscosity of 1.8 Pa.s were jetted [92].

2.2. Hybrid-actuation mode

Fig. 9(a) shows the schematic of the EHD assistant type PMJD, which generates the DODJ ultra-fine droplets easily based on the EHD actuation and the piezoelectric actuation. The piezoelectric actuation mode can deform the fluid meniscus and extrude the outside of capillary, while the EHD actuation can realize ultra-fine droplets because the electric potential on the surface of meniscus exceeds a critical value due to its high voltage applied between the capillary and pin type electrode. A few of researches on the EHD assistant type PMJDs have been done [94–101]. Kim et al. proposed a novel PMJD under the EHD and piezoelectric force to generate ultra-fine droplets, and the droplet volumes were from

3.4 to 46.8 pL [94,95]. Luo et al. proposed a EHD assistant type PMJD to jet the repeatable droplet with 600 pL [96]. Chen et al. proposed the novel method for single cell analysis [97], as seen in Fig. 9(b). Xu et al. studied the droplet dynamic behavior based on the EHD assistant PMJD, and the droplet diameter was significantly reduced to less than 1/5 of the nozzle diameter [98]. Yu et al. presented an improved probe cell mass spectrometry method using the EHD assistant type PMJD, which had become a powerful tool for single cell analysis due to its low cost, easy to handle and multi-function [99].

Table 3 shows the performance comparisons for the PMJDs. Firstly, for the shear type PMJDs [41–44], the companies such as Xaar and Spectra are the pioneers [42]. The merit of the shear type PMJDs is that the shared-wall principle, where the piezoelectric elements are also the fluid channel, is adopted to facilitate the miniaturization design. The shear type PMJDs are widely used for inkjet printers in home and office. However, the production cost of the shear type PMJDs is extremely expensive due to the complex fabrication, and the cross-talk problem cannot be avoided because every third adjacent channel is driven during operation process. Thus, their engineering applications are limited severely.

Compared to the shear type PMJDs, the simple structure designs for the squeeze type PMJDs are realized and the various materials can be distributed easily [45–54]. However, a larger acoustic impedance of the fluid flow caused by the thin and long channel may result in energy loss. In addition, the air bubble and the channel fluid reflux issues which affect the droplet behavior and even

Table 3
Summary of properties for the PMJDs.

Reference	Years	Actuation mode	Nozzles Diameter (μm)	Frequency (Hz)	Voltage (V)	Droplet Diameter (μm)	Droplet volume (nL)	Droplet Velocity (m/s)	Viscosity (cps)
[44]	2005	Shear	4	NA	140	3.35	NA	NA	NA
[49]	2001	Squeeze	70	10000	70	70	NA	3	15.0
[50]	2001	Squeeze	50	50	NA	NA	0.0006	1.8	1.0
[52]	2010	Squeeze	119	120	50	150	NA	3.2	100.0
[53]	2011	Squeeze	50	200	230	63	NA	NA	14.3
[58]	1999	Bend	60	500	150	40	0.035	NA	NA
[61]	2008	Bend	300	40	80	NA	160	1.4	17.4
[66]	2014	Bend	307	250	245	450	NA	0.3	NA
[67]	2016	Bend	100	1830	100	186	NA	8.4	8.9
[68]	2017	Bend	100	160	200	NA	NA	16.8	1.0
[72]	2006	Push	53	20	21.6	39	NA	8.2	5.0
[73]	2007	Push	50	100	NA	82	0.288	4.5	NA
[75]	2012	Push	100	4126	200	NA	NA	82.6	430
[77]	2008	Needle	250	500	70	NA	NA	3.7	10000
[82]	2013	Needle	250	65	120	1070	NA	NA	900
[83]	2018	Needle	200	400	180	525	NA	NA	180
[85]	2015	Needle	100	NA	100	NA	620	NA	58000
[86]	2017	Needle	200	150	100	NA	25	NA	1400
[95]	2013	EHD	50	2000	27	NA	0.0029	NA	8.9
[96]	2013	EHD	55	NA	3500	NA	0.6	NA	1.8
					56				
[98]	2014	EHD	110	50	3500	20	NA	NA	1.0
					40				
					5000				

cause the jetting failure are prone to occur. Similar to the squeeze type PMJDs, the bend type PMJDs are simple structure and low cost [55–69], which are widely applied to biomedicine, material chemistry, 3D printing technology and space mechanism lubrication. However, the residual vibration and air bubble are hardly avoided, which influence the droplet uniformity and productivity. For the push type PMJDs [70–76], their structures are similar to the bend type PMJDs. The difference between them lies in the installation method of piezoelectric elements. The superiority of the push type PMJDs is that the piezoelectric element doesn't contact with the fluid directly, which can avoid the undesirable chemical change between the liquid and piezoelectric element effectively. Meanwhile, the push type PMJDs are easier to realize the high-temperature fluid jetting process.

Four PMJDs mentioned above can only be used to jet the low viscosity fluid. In recent years, the precise distribution of the high viscosity fluid such as adhesives and silver paste is an important guarantee to ensure the quality of electronic packaging. Some researches on the needle-collision type PMJDs are proposed to jet the high viscosity fluid [77–93]. However, the jetting frequency is severely limited, and the structure configuration is so complex that it is difficult to be miniaturized. In addition, to obtain an ideal droplet performance, a smaller channel structure is designed to restrain the channel fluid reflux, and a smaller nozzle size is also generally fabricated. But the manufacturing difficulty and the cost are increased obviously.

On the basis of the unchanged structure, an ideal droplet can also be obtained by hybrid actuation mode [94–101], which is more suitable for dispensing the droplet by taking on a conical profile, a Taylor cone and a fine jet. Although the EHD assistant type PMJD needs to place the stringent requirements on both the physical properties and operating conditions, it can generate a micron-size droplet from a millimeter-size tube easily.

3. Analysis methods of PMJDs

The analysis methods involve the coupling of structural, electric and interfacial flow fields, which can be realized by direct coupled-field simulation [102–104]. But the direct coupling method based

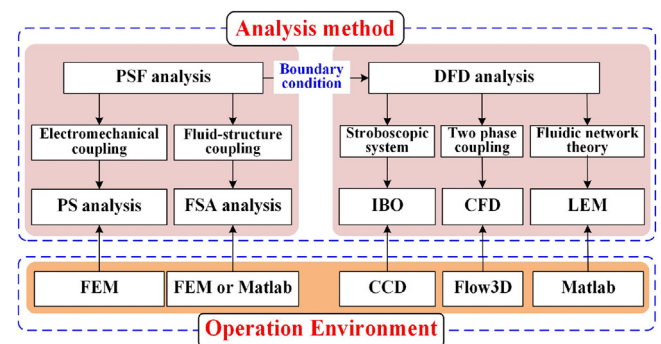


Fig. 10. Flow chart of the indirect coupling method of the PMJDs.

on multi-physical fields requires substantial calculation power and time cost. An indirect coupling method, so-called load transfer, coupling multi-physical fields to apply the results from one analysis as the load in another analysis, has been proved to be more effective to analyze the PMJDs. The indirect coupling method is shown in Fig. 10.

The indirect coupling method includes piezoelectric-structure-fluid (PSF) analysis and droplet formation dynamic (DFD) analysis, while the PSF analysis method includes piezoelectric-structure (PS) analysis which is achieved by finite element method (FEM) and fluid-structure acoustic (FSA) analysis which is realized by FEM and Matlab based on the narrow gap theory.

3.1. Piezoelectric-structure-fluid analysis

Most contributions of PS analysis on the PMJDs are to optimize the design parameter of transducer and to determine the resonant frequency, which can effectively guide the structure design and experiment analysis. The optimum radius and thickness of piezoelectric element were analyzed (eg., Peelamedu et al. [105], Morris et al. [106] and Chang et al. [107]). The optimum membrane width and diameter were analyzed by Nguyen et al. [108] and the effect of the resonant frequency on displacement response was also studied by Liu et al. [109]. However, the coupling interaction between

vibrator and fluid is ignored. Although the coupling interaction doesn't affect the modal shapes, the resonance frequencies of the corresponding modal are significantly reduced.

Considering the effect of the fluid-structure interaction, many relevant researches on FSA analysis method have been carried out [110–125]. The pulse frequency (resonant frequency) was determined accurately by the FSA analysis by Teng et al. [110,111]. The dynamic characteristics of the viscous and incompressible fluid were predicted using the FSA analysis method by Dijkman et al. [112] and Shin et al. [115]. The influences of the fluid physical properties and shell materials on nozzle pressure and support reaction were analyzed by Liu et al. [123], and the effects of the structure parameters on the acoustic pressure wave and droplet velocity were also simulated [124]. By the PSF analysis method, the fluid pressure and velocity at the nozzle are obtained, which can be transmitted to the DFD analysis as the boundary conditions to obtain the droplet characteristics.

3.2. Droplet formation dynamic analysis

The DFD analysis includes imaging-based observation (IBO) [126–136], computational fluid dynamics (CFD) [137–157] and lumped element model (LEM) [158–163]. Firstly, the IBO method can be used to study the droplet dynamic performances, which is recorded by a CCD camera with a strobed LED, a high-speed camera, a laser-induced fluorescence illumination (eg., Attinger et al. [126], Heij et al. [127], Furbank et al. [128], Meacham et al. [129], Tsai et al. [130–132] and Bos et al. [135]). The principle of the IBO is that the region is illuminated with a pulsed light, in which the fluid emerges in the jetting form and then forms into the droplets. Besides, the edge detection was reported by Kwon et al. [133], and the error rate of the droplet diameter could be controlled at about 3%. Compared with the traditional algorithm, the edge detection algorithm had the merits of more intuitive, convenient and effective results. In fact, the chamber is generally non-transparent and the sensor is not easy to be placed inside the chamber. Thus, it is limited to study the droplet characteristics.

The numerical method is also used to analyze the dynamic behavior of the droplet forming process. Early models failed to predict the temporal evolution of the droplet in velocity, shape and trajectory aspects due to the inadequate interfacial physics models and the complicated topology variations. As the rapid development of the numerical method and computing power, the CFD has become a promising tool to overcome the limitations of theoretical models, which is realized using the VOF code Flow3D that includes wall-flexibility and free surface flow with surface tension. Relevant researches on droplet formation process, profile, velocity and volume were investigated based on the CFD (eg., Feng et al. [137], Wu et al. [138–141], Yang et al. [142], Kim et al. [146], Liou et al. [147], Yao et al. [151]). The structural parameters such as nozzle shape and chamber size were also analyzed by Suh et al. [144] and Liu et al. [149]. In addition, the effects of pulse signal on droplet volume and velocity were studied by Desai et al. [145] and Lai et al. [148]. The spreading performance was also analyzed by Liu et al. [152–154]. But the CFD based on mesh method is extremely time-consuming. As an alternative method of the CFD, the LEM is constructed to study the droplet dynamics based on fluidic network theory. Moreover, it is similar to the circuit simulation which can be realized by an equivalent circuit and fluid dynamics equations. Although the calculation accuracy of the LEM is limited due to the assumptions, a much lower computational burden makes it more promising for some comprehensive optimization on the PMJDs.

The LEM method can quickly predict the basic performances of the PMJDs such as frequency response in terms of diaphragm displacement, chamber pressure, droplet velocity and volume, which is achieved based on equivalent circuit [158–160] and fluid dynam-

ics equations [161–163]. The individual components were modeled as elements of the equivalent circuit (eg., Kim et al. [158], Lu et al. [159] and Ono et al. [160]), and the droplet volume and velocity could be accurately predicted. Besides, the LEM based on fluid dynamics equations was also presented by Luca et al. [161,162], which represented a practical tool for overall design of the PMJDs. The comparisons of the LEM method between the equivalent circuit and fluid dynamics equations were well presented in a previous review by Chiatto et al. [163] and their useful links also were discussed.

4. Challenges of PMJDs

Although the wanted droplet performances can be realized by proper actuation mode, structure configuration and analysis method, there are still many studies to be completed and optimized to solve the bottlenecks faced by the PMJDs, such as low droplet uniformity, low droplet productivity and limited jetting object.

4.1. Low droplet uniformity

The factors that affecting the droplet uniformity are mainly attributed to the air bubble entrapment, cross-talk effects and changing dynamics.

4.1.1. Air bubble entrapment

The air bubble entrapped at the nozzle is a key factor to degrade the droplet uniformity. Once entrapped, the air bubble grows by rectified diffusion, and results in the malfunction. The air entrapments include two aspects: for one thing, the asymmetric meniscus is formed at the nozzle due to small dust particles, and the asymmetry structure of the retracting meniscus in combination with the next symmetric pressure wave may cause the air bubble entrapment; for another, the fluid layer on the nozzle plate may also be created, which may favor the void formation once the meniscus is pulled back and the fluid closes the void.

4.1.2. Cross-talk effect

The actuation of one channel affects the neighboring channels, which is called cross-talk effect including the electrical, acoustic and structural cross-talk. In the electrical cross-talk, the piezoelectric element is influenced when the neighbor is actuated. However, the electrical cross-talk effect is very slight. For the acoustic cross-talk effect, the pressure wave propagates in a channel and influences the other channels. It is also assumed that it has a marginal effect on the PMJDs. The structural cross-talk effect including direct and indirect cross-talk effect is inevitable because one channel deformation leads to neighboring channel deformation. The deviation of droplet velocity and size, deflection of droplet angle and unstable droplet behavior may be produced.

4.1.3. Changing dynamics

The same pulse signal cannot always generate the same droplet, due to fluid degradation, aging of the PMJDs, environmental changing such as temperature, nozzle surface wetting and blockage of the nozzle.

4.2. Low droplet productivity

The degradation of the droplet productivity is mainly attributed to residual vibration, channel fluid reflux and satellite droplet generation.

4.2.1. Residual vibration

The actuation energy of the chamber generated by acoustic pressure wave is propagated from the channel to the nozzle, and the

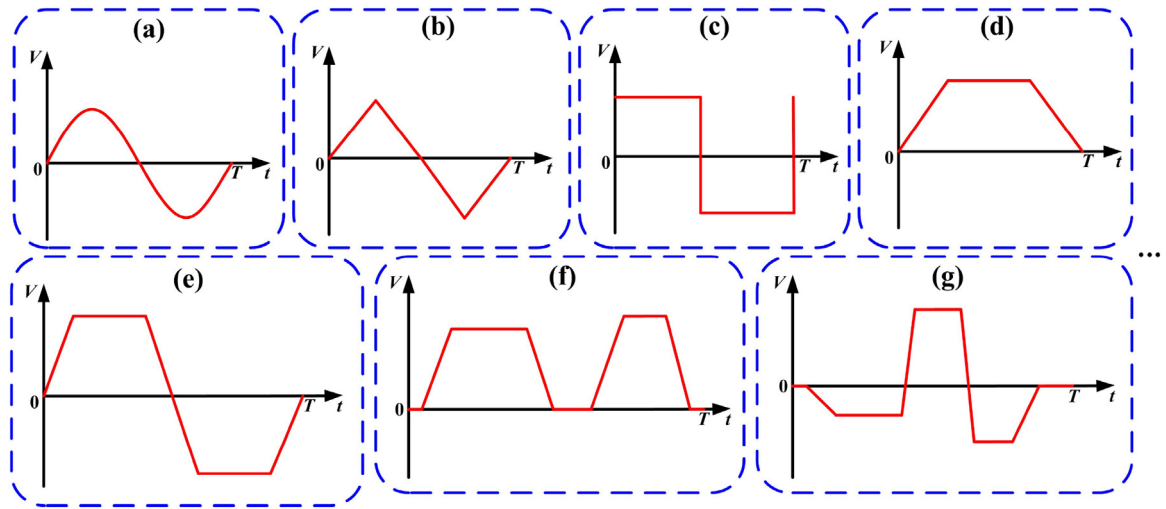


Fig. 11. Typical pulse waveform of the PMJDs: (a) sinusoidal pulse; (b) triangular pulse; (c) trapezoidal pulse; (d) square pulse; (e) bipolar pulse; (f) M-shaped pulse; (g) W-shaped pulse.

droplet can be jetted at the nozzle. After jetting the droplet, the fluid cannot be at rest immediately due to residual vibration. To ensure the droplet uniformity and reliability, the channel must be actuated at rest condition. However, the required time for the residual vibration to damp out is larger than a short operation time. Generally, the fluid channel needs to take around $200\ \mu\text{s}$ to rest such that the next droplet can be jetted well [211]. Although the droplet properties can be ensured when the channel is actuated each time, the droplet productivity is seriously reduced.

4.2.2. Channel fluid reflux

The channel fluid reflux is also an important factor affecting the droplet productivity. When the piezoelectric element is actuated, one part of fluid enters the chamber along one direction of the channel and is jetted from the nozzle, while the other part of fluid flows to the inlet along another direction of the channel, which is named as channel fluid reflux. The channel fluid reflux reduces the droplet productivity, and even leads to malfunction. Currently, the channel fluid reflux issue can be restrained by decreasing channel size. However, this method greatly increases the manufacturing difficulty and the production cost.

4.2.3. Satellite droplet generation

The satellite droplet cannot be generated at a lower droplet velocity but tend to gradually appear as the droplet velocity increases. To ensure the image quality, the droplet productivity is reduced seriously. It is necessary for the PMJD to find a novel method to achieve a high-speed satellite-free operation.

4.3. Limited jetting object

The factors that limiting the fluid jetting object are mainly attributed to a low Curie temperature and a small actuation displacement of piezoelectric element.

4.3.1. Low Curie temperature

A higher fluid temperature will make the PMJDs work invalid because of a lower Curie temperature of piezoelectric element. Thus, it is difficult to jet a normal droplet under a higher temperature condition.

4.3.2. Small actuation displacement

The droplet can be jetted by overcoming the shear stress between the nozzle inner wall and the fluid. However, for the high

viscosity fluid, the shear stress is too large to achieve a successful jetting operation because of a small actuation displacement. Recently, some needle-collision type PMJDs were reported to jet the high viscosity fluid. However, the jetting frequency of the PMJDs is severely limited.

5. Uniformity and productivity improvements of PMJDs

With the rapid development of additive manufacturing technology, the droplet characteristics are quite tight. A higher requirement for droplet uniformity and productivity are frequently imposed. It is expected that the performance standard becomes even tighter to motivate the relevant research into the PMJDs.

5.1. Structural optimization design

The structural optimization of the PMJDs mainly focus on shell material, chamber structure, nozzle size and nozzle layout. The influences of shell material and chamber structure on droplet velocity were researched by Liu et al. [123,124]. The relationships between nozzle diameter and droplet volume were also studied (eg., Ozawa et al. [164], Chen et al. [165], Liou et al. [166], Iwanaga et al. [167] and Minov et al. [168]), and the femtoliter droplet was obtained by reducing the nozzle diameter. The effect of nozzles layout on droplet performances was studied by Feng et al. [169], and the reasonable nozzle layout was achieved by overlapping the nozzle with amplitude contour map of the pressure wave. It can be seen that the nozzle diameter is generally reduced to generate an ideal droplet. However, a smaller nozzle size is much difficult to fabricate and easy to be blocked. Alternatively, the pulse signals are designed to generate ideal droplets without changing the structure.

5.2. Waveform design and optimization

The typical pulse signals of the PMJDs include sinusoidal, triangular, trapezoidal and square waveform, as shown in Fig. 11(a–d). Among them, the trapezoidal waveform is the most widely used in the PMJDs, which is also named as standard pulse. But the residual vibration is generated using the standard pulse. To suppress the residual vibration, some novel waveforms including bipolar, M-shaped and W-shaped waveform are proposed by adding the additional pulse signal on the basis of the standard pulse, as shown in Fig. 11(e–g).

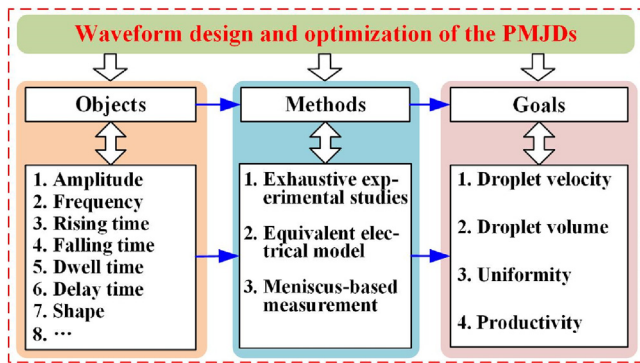


Fig. 12. Flow chart of waveform design and optimization of the PMJDs.

Fig. 12 shows the waveform design and optimization method for the PMJDs, and the final goals such as droplet velocity, volume, uniformity and productivity can be achieved by exhaustive experimental studies [170–180], equivalent electrical model [181–185] and meniscus-based measurement [186–189]. The optimization objects of the pulse signals mainly consist of the amplitude, frequency, dwell time and shape.

Firstly, the parameter optimization of pulse waveform is generally designed by exhaustive experimental studies. The shapes of the pulse waveform, unipolar waveform, bipolar waveform, M-shaped and W-shaped waveform, were designed and optimized to generate a smaller droplet volume and a larger droplet velocity (eg., Szczech et al. [170], Wu et al. [171,172] and Gan et al. [173]). The influences of pulse voltage and frequency on droplet uniformity and productivity are researched (eg., Beulen et al. [174], Saunders et al. [175] and Li et al. [176]). The rising, falling, dwell and delay time were studied by Kwon et al. [177,178], and the satellite droplet was suppressed by the 20–30% longer dwell time with respect to the efficient dwell time. Besides, the droplet diameter could also be reduced by reducing the duration time to form the fine line pattern on the polyimide films. Although the pulse waveform can be redesigned and optimized by the exhaustive experimental studies, this method demands a precise mechanical alignment of the sensor and the corresponding device. So it is time-consuming to obtain an ideal waveform design result, and it is much necessary to improve the design efficiency of the pulse signal.

The number of experiments for designing the pulse waveform can be reduced by reasonable guess on the pulse parameters. The pulse parameters are closely related to fundamental period of channel pressure wave which can be obtained by using an equivalent electrical model based on piezoelectric self-sensing method [181,182] and LEM method [183–185], and the pulse waveform can be rapidly designed and optimized. In addition, the design parameters are also closely related to the vibration period of the pressure wave, and the vibration period can be obtained based on meniscus motion measurement.

The meniscus is the interface of the fluid and the air in the nozzle. Once the fundamental period of meniscus motion is measured, the pulse waveform can be designed and optimized. Kwon et al. proposed a waveform design and optimization method, and the optimum dwell time was determined by meniscus motion period. [186]. Shin et al. utilized the meniscus motion to design a double waveform [187]. Wei et al. studied the rise, descent and dwell time of unipolar waveform to acquire the optimal waveform [188]. Xiao et al. optimized the pulse width combining meniscus motion measurement and numerical simulation [189], which reduced the experimental efforts and saved lots of time cost to determine the optimal pulse parameter relative to exhaustive experimental studies. In addition, the designed waveform can be evaluated without

the need for the actual jetting condition since the jetting behavior can be predicted by measuring the meniscus motion.

5.3. Fault detecting technology

The air bubbles are known to be a major cause of malfunction due to the poor stability. To avoid this malfunction, any problems in the jetting process must be identified and solved immediately. Besides, the other malfunctions include the changing dynamics such as fluid degradation, aging, environmental temperature and nozzle blockage. Relevant researches have been done to monitor the droplet conditions, such as vision-based method [190–197], piezoelectric self-sensing method [198–206] and meniscus motion monitor [207–209].

The vision-based method is the most common method for detecting the droplet behavior due to its simplicity and versatility. The detection of the air bubble and droplet flying process has been done (eg., Kim et al. [191], Fan et al. [192], Bos et al. [193], Hoath et al. [194] and Kwon et al. [195,196]). The inspection of 384 fictitious droplets was performed in 17 s with accuracies of $0.2 \pm 0.5 \mu\text{m}$ for a $50 \mu\text{m}$ droplet and $1.2 \pm 0.3 \mu\text{m}$ for a $30 \mu\text{m}$ droplet [197]. However, the vision-based method needs to take an amount of time to scan all nozzles when monitoring a multi-nozzle device, and it is difficult to detect and analyze the possible malfunction immediately. In addition, the malfunction cannot be detected utilizing the vision-based method when no droplet is jetted.

The piezoelectric self-sensing method has been a promising alternative for the vision-based method. For one thing, the piezoelectric actuator can be used to drive pressure chamber to generate the droplets; for another, it can also be used as a sensor by sensing mechanical vibration. Jong et al. firstly investigated the air bubbles using piezoelectric element by monitoring acoustic signals [198], and the air bubble influenced the acoustic signals and reduced the pressure wave at the nozzle, in which was insufficient for jetting the droplets [199–201]. The self-sensing method has been gradually developed (eg., Jeurissen et al. [202], Kim et al. [203,204], Kwon et al. [205,206]). The different self-sensing behaviors could be obtained for various malfunctions, such as unsuitable temperature control, abnormal backpressure and nozzle blockage. Although the air bubbles was monitored well using self-sensing method, the actual jetting condition at the nozzle is difficult to be judged because the dynamic behaviors of pressure wave were measured at the location of piezoelectric element rather than the nozzle.

The limitation of the self-sensing method can be overcome by measuring the meniscus motion at the nozzle effectively. Kwon et al. monitored the jetting conditions and detected the possible malfunction based on the meniscus motion [207]. Wei et al. monitored the meniscus motion during the droplet jetting process, and the meniscus motion was used for in-situ diagnosis [208]. Nguyen et al. improved the droplet uniformity and productivity by stabilizing the meniscus shape based on the visual feedback algorithm [209].

The advantages and possible limitations of malfunction detection are also discussed in the PMJDs. Table 4 shows the performances among above methods for malfunction detection. Compared with the vision-based method and the meniscus motion method, the self-sensing method has the high efficiency since this method only needs to measure the electrical signals, and the condition monitoring was possible at any place as well as at any time. However, the actual jetting at the nozzle cannot be predictable by the self-sensing method. Compared with the self-sensing method, the meniscus motion method can predict the droplet behavior since it is measured at the nozzle where the actual droplet jetting occurred.

Table 4

Performance comparisons of the above methods for malfunction detection.

No.	Malfunction type	Vision-based	Self-sensing	Meniscus motion
(1)	Detection speed	Very slow	Very fast	Slow
(2)	Malfunction detection during the printing process	Impossible	Possible	Impossible
(3)	Malfunction detection during jetting or non-jetting	Jetting	Both	Non-jetting
(4)	Blockage at nozzle	Undetectable	Undetectable	Detectable
(5)	Jetting prediction	Impossible	Difficult	Very good
(6)	Measurement of negative pressure wave	Non-measurable	Measurable	Non-measurable
(7)	Measurement of positive pressure wave	Non-measurable	Measurable	Measurable

5.4. Control strategy

A small droplet size with high frequency is limited due to residual vibration and channel cross-talk effect. To overcome these two disadvantages, Zhan et al. designed a fuzzy PID control system to obtain a softer and cleaner image [210]. A feed-forward method was proposed to restrain the residual vibration without influencing the droplet formation [211–213] and this method was also optimized by Khalate et al. [214–218]. In addition, the residual vibration and cross-talk effect could also be removed by the iterative learning control (eg., Oosten et al. [219], Wassink et al. [220,221] and Velden et al. [222]), and the droplet velocity and volume variations were reduced by 52% and 41% in terms of the standard deviation. A novel closed-loop control method based on adaptive wavelet neural network controller was proposed, and this method not only outperformed the PID control but also worked well in the performance optimization [223].

5.5. Fluid characteristics analysis

The dispensing technology based on the PMJDs is a rapidly growing commercial process for applications depending on precisely jetting micro-scale droplets. These applications increasingly require the complex fluids involving the viscoelastic properties. It is essential to get a clear comprehension of the relationship between the droplet dynamics and the fluid physical properties which are defined by Reynolds number (Re), Weber number (We) and Ohnesorge number (Oh). These dimensionless numbers are defined by the following equations:

$$Re = va\rho/\eta \quad (1)$$

$$We = v^2a\rho/\gamma \quad (2)$$

$$Oh = \sqrt{We}/Re = \eta/\sqrt{a\rho\gamma} \quad (3)$$

where ρ , η and γ represent the density, dynamic viscosity and surface tension; v is the droplet velocity and a is the nozzle radius. Another dimensionless is inverse (Z) of the Oh , which is independent on fluid velocity:

$$Z = 1/Oh = \sqrt{a\rho\gamma}/\eta \quad (4)$$

Newtonian fluid was used and studied widely in the droplet formation starting back with Rayleigh in the nineteenth century. Later, relevant researches have been done one after another [224–229]. The fluid used as the functional materials must be printable and stable. Jo et al. studied the effects of the viscosities of the Newtonian liquids on the droplet characteristics [224]. Derby analyzed the influence of the Z on the droplet characteristics and the printable range was within $1 < Z < 10$ [226], as shown in Fig. 13.

When the Newtonian fluid is jetted from a nozzle, a small satellite droplet is generated along with a primary droplet and degrades

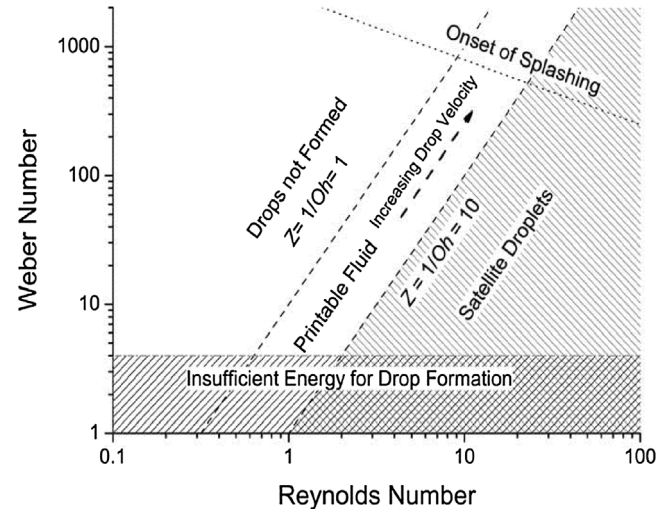


Fig. 13. The printable range of the Newtonian fluid in a parameter space defined by the axes of Reynolds and Weber numbers [226].

the printing resolution. To restrain the satellite droplet, the influences of dimensionless numbers on droplet dynamic behaviors were investigated by Kim et al. [227] and Poozesh et al. [228,229]. In addition, the rheological properties of polymer solutions are highly nonlinear. Thus, the rheological properties cannot be ignored to produce an ideal droplet. Relevant researches on complex fluid have been done [230–241]. Shore et al. studied the effect of the viscoelasticity of low viscosity fluid on droplet formation, and the satellite droplet was restrained by adding the 10–50 ppm high molecular polymer [231]. Castrejón-Pita et al. studied the effects of the Re and We number on droplet behavior, and the droplet velocity range of 0.3–6.0 m/s was realized [232]. Matavz et al. studied the droplet shape and drying mechanism under various viscosities and temperature conditions [235]. Yoo et al. studied the droplet formation mechanisms of rheologically complex fluids [238] and the dynamic characteristics from droplet generation to complete drying [239]. Bienia et al. studied the filament growth and breakup in printing process of the ceramic suspensions [240]. Cheng et al. achieved the reliable cell dispensing by increasing the viscosity and density of the cell suspension [241].

6. Applications of PMJDs

The PMJDs are viewed as a versatile manufacturing tool in biomedicine, material chemistry, microelectronic device, 3D printing, packaging technology and space mechanism lubrication [242–246], due to their non-contact, additive patterning and mask-

less, low cost, environmental friendly, rapid dynamic response, high precision.

6.1. Biomedical application

The applications of the PMJDs in biomedical field have been extensively studied by domestic and foreign researchers [247–267]. Hasenbank et al. created a sub-millimeter pattern of the aqueous and non-aqueous reagent for biosensor [249]. Doraiswamy et al. controlled the flow of the mussel adhesive protein solution to achieve wound repair, fracture fixation, wound closure and drug delivery [251]. Scoutaris et al. distributed the mixture of the drug felodipine, antihypertensive and polyvinyl pyrrolidone to the hydrophobic substrates [252]. Lee et al. produced the drug-loaded polymer microparticles with the well-defined and controlled shapes [253]. Chahal et al. printed the breast cancer cells, and the accuracy and reliability were enhanced by neutral buoyancy of the cell suspension [254]. Chen et al. proposed a novel chemiluminescence diagnosis system for human IgA detection [255], as seen in Fig. 14(a). Fujita et al. created a super-dense transfected cell microarray (TCM) on a glass substrate that had been grafted with the poly (ethylene glycol) [256], as seen in Fig. 14(b). Boehm et al. added the amphotericin B on the biodegradable microneedles which were used for transdermal delivery of the pharmacologic agents [257]. Lorber et al. printed the adult rat central nervous cell, retinal ganglion cells and glia [259]. Boehm distributed antifungal agent and voriconazole to the surfaces of biodegradable polyglycolic acid microneedles [261]. Spears et al. envisioned an ideal fabrication technique to provide the particle with a controlled size, which allowed for printing the polymeric structure onto a substrate [265], as seen in Fig. 14(c).

6.2. Material chemistry

The material chemistry applications of the PMJDs have been researched all over the world [268–280]. Lejeune et al. fabricated the different kinds of micro-pillar array structures based on PZT and TiO₂ suspensions [271], as seen in Fig. 15(a). Zeng et al. developed a PMJD for capillary electrophoresis capable of reproducing a picoliter volume [273]. Hanuš et al. printed the alginate-liposome-magnetite microparticles [274]. Kadota et al. produced the monodispersed titania particles for sol-gel processing [276], as shown in Fig. 15(b). Staymates et al. produced the discrete microparticles from compounds dissolved in nonpolar or polar solvents, and the ammonium nitrate particles were fabricated [277]. Kuscer et al. produced a lead-zirconate-titanate thick-film by printing aqueous zirconate-titanate-powder and polyethyleneimine solution powder on a metalized alumina substrate [278]. Salaoru et al. produced the polyvinyl alcohol (PVOH) multilayer structures, and the morphology, surface profile and uniformity of the multilayers were evaluated by optical and FT-IR microscopy [280].

6.3. Microelectronic device

Many researches on microelectronic devices have been done [281–296]. Redinger et al. created a high quality passive device for RFID front-end [281]. Koo et al. fabricated a greenish color filter layer for active-matrix liquid crystal display [283,284]. Sridhar et al. printed a silver nanoparticle on the reinforced substrate with the available fiberglass material [286]. Lee et al. fabricated the conductive patterns on a flexible polyimide film based on the Ag nanoplate ink of the 10% solid content [287], as shown in Fig. 16(a). Li et al. printed the resistor and capacitor based on two print-heads, which overcame the limitations of single print-head system in various fabrications [288]. Chiolerio et al. fabricated the low-frequency and high-frequency negative capacitors, and the highest negative

supercapacitance was -2.3 mF at 30 Hz [291]. Cinti et al. studied a sensor for the simple and sensitive measurement of the hydrogen peroxide based on the printed electrodes modified with Prussian blue nanoparticles [292]. Ning et al. fabricated the thin-film transistors (TFTs) by printing the silver (Ag) source/drain electrodes on the amorphous indium-gallium-zinc-oxide (a-IGZO) layer [294,295], as seen in Fig. 16(b). Ha et al. fabricated the organic thin-film transistors by printing the conductive metal-organic precursor type silver liquid [296].

6.4. 3D printing technology

3D printing technology mainly includes fused deposition [297,298], selective laser melting/sintering [299,300] and PMJDs. Compared to the fused deposition and selective laser melting/sintering, the PMJDs are becoming more and more prominent in the 3D printing technology. At present, the PMJDs are mainly used to print the low viscosity fluid at room temperature such as biology and medicine applications [301–316]. Uddin et al. printed the 5-fluorouracil, curcumin and cisplatin on a coat metal microneedle array accurately for transdermal delivery [303], as seen in Fig. 17(a). Amorn et al. printed ten different functional materials based on a multi-material 3D printing platform with high resolution, low-cost and extensibility [304]. Zhang et al. printed the multiple cells within biocompatible materials for microfluidic chips [305]. Gao et al. fabricated the alginate microparticles into the different shapes [308].

A few researches on the metal 3D printing have also been done. Faddoul et al. printed a ceramic substrate by indirect processing of the silver nanoparticles. The metal pattern was obtained by sintering process [312]. Wang et al. proposed a 3D printing system of molten lead-free solder [313], as seen in Fig. 17(b). Ma et al. produced the droplets with about $85 \mu\text{m}$ and serve as an alternative method for the 3D metal printing [314]. Qi et al. printed the high temperature molten metal materials [315,316], as seen in Fig. 18(c). Chen et al. fabricated a conductive copper pattern on flexible polyimide substrate and two-step electroless plating procedure, which performed good solderability, reliable adhesion strength and low resistivity of $5.68 \mu\Omega \text{ cm}$ without any sintering process [317].

In summary, the piezoelectric 3D printing is generally oriented to normal or medium-low temperature fluid. However, the printing process of high temperature molten metal materials is limited because of a lower Curie temperature. It can be predicted that if the piezoelectric 3D printing technology can break through the limit in the high temperature and high viscosity fluid, it will not only solve the bottlenecks of the integrated forming of complex structural parts in military industries such as aerospace but also be popularized in civilian industries.

6.5. Packaging technology

The microelectronic packaging technology is composed of a contact and non-contact dispensing. The non-contact method has been studied relative to the contact dispensing. Nguyen et al. designed a piezo stack type PMJD that was applied to semiconductor packaging [77,78]. Wang et al. designed a PMJD based on the corner-filled flexure hinge for high-frequency microelectronic packaging, and a smaller droplet diameter of $525 \mu\text{m}$ was steadily obtained with a high frequency of 400 Hz [83]. Liu et al. used the needle-collision type PMJD to dispense the industrial viscosity of 58 Pa.s in glue spray and the volume of $0.62 \mu\text{l}$, which could be applied to electronic packaging [85]. Zhou et al. proposed a novel high-speed PMJD based on double piezo stacks to distribute the adhesives rapidly for microelectronic packaging [86,92].

In a word, the precise distribution of high viscous liquid materials such as adhesives and silver paste is an important guarantee

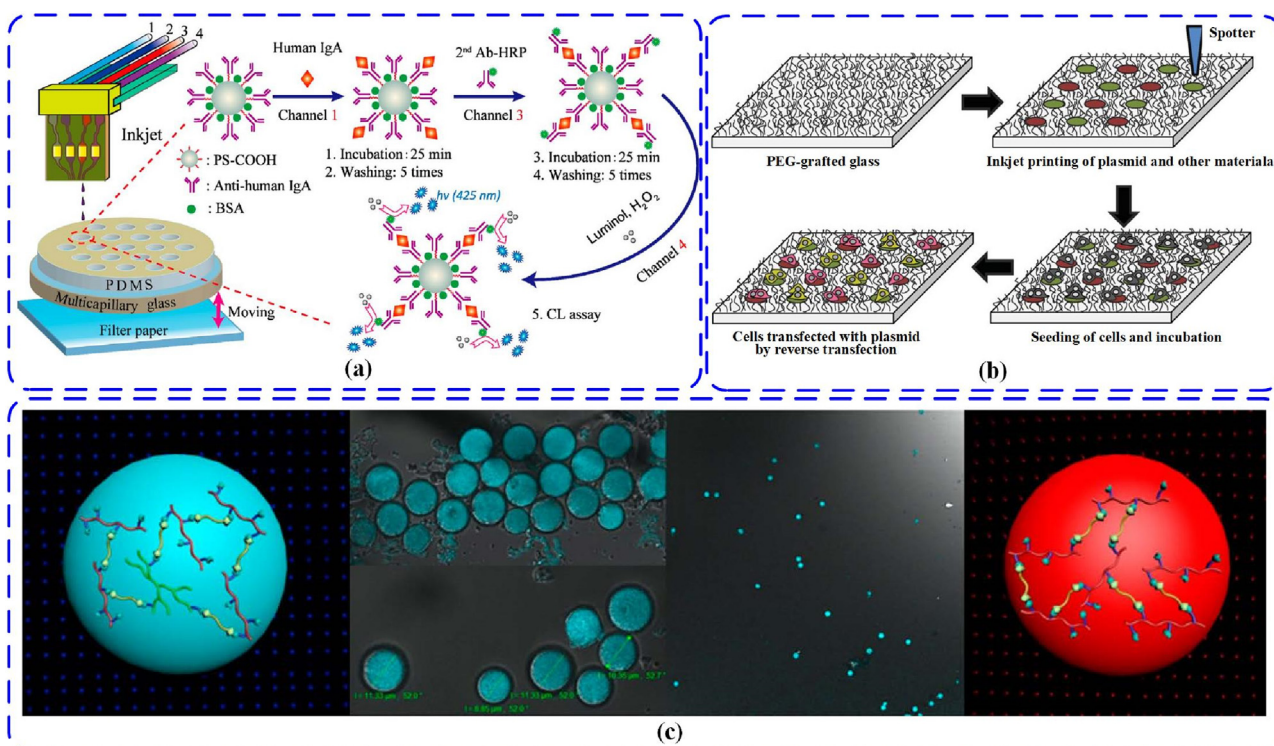


Fig. 14. Biomedical applications of the PMJDs: (a) schematic illustration of the sandwich chemiluminescence immunoassay for human IgA detection [255]; (b) schematic illustration of the construction of super-dense transfected cell microarray [256]; (c) structures of the precise microscale polymeric networks [265].

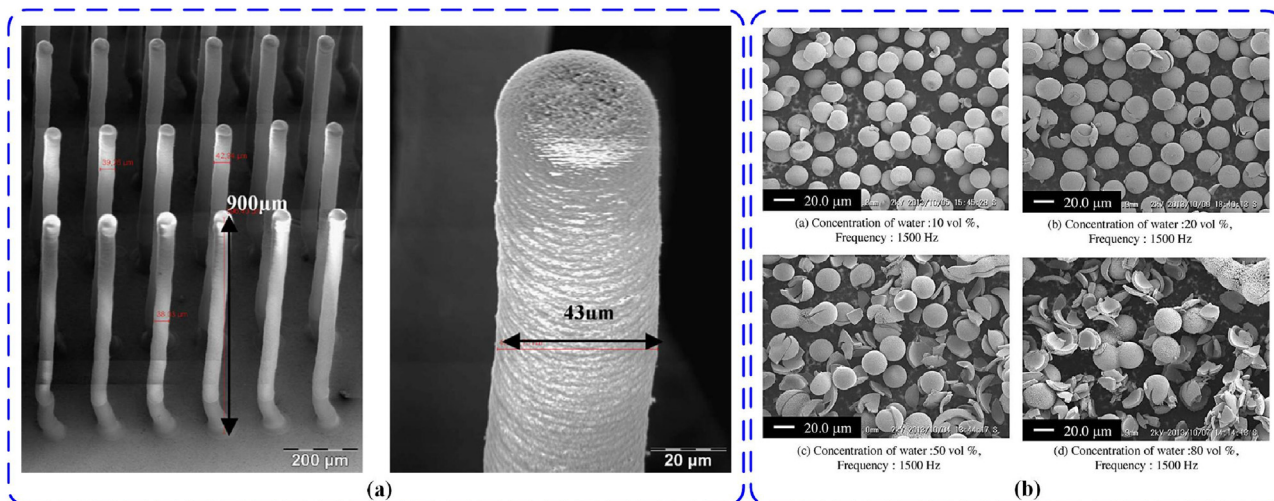


Fig. 15. Material chemistry applications of the PMJDs: (a) scanning electronic images of the green PZT micro-pillar array fabricated on the PZT green tape [271]; (b) SEM images of the titania particles prepared with various water concentration [276] (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article).

to electronic packaging. It is a pity that the precise distribution of the high viscous liquid materials has always been a technical bottleneck, which restricts the development of electronic packaging towards a higher productivity, miniaturization and three-dimensional scale.

6.6. Space mechanism lubrication

In space mechanism application, the liquid lubrication is generally used to provide the reliable and long-term performance for many moving mechanical assemblies. During the past decades, the mission lifetime was short due to the failing of electric and

computer system before the lubrication system. With the rapid advance of the reliability and lifetime of electric and computer system in recent years, the reliability and lifetime have gradually turned to the issue of how to prolong the reliability and lifetime for moving parts of space mechanism. Limited by high vacuum, high-low temperature and high radiation, the lubrication system has become the main factor in determining the reliability for space mechanism [318]. In many cases, the premature failure of the lubrication system can result in mission failure, and the most lubrication technologies belong to passive lubrication [319]. But the issues of the passive lubrications are whether the lubrication is timely and the service life can reach more than ten years. Based on this,

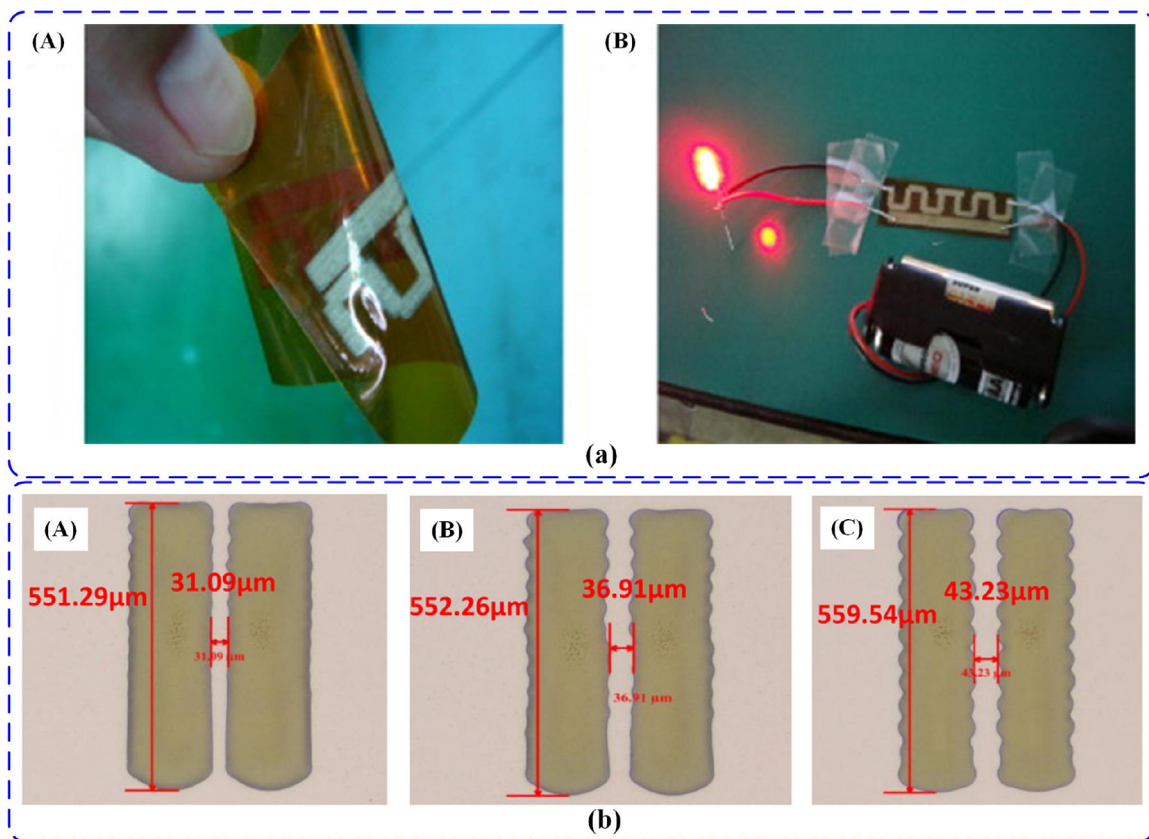


Fig. 16. Microelectronic device applications of the PMJDs; (a) photographs of pattern printed on a PI film by carrying out the printing-baking five times: (A) rolling-up and (B) feasibility of the conductive circuit [287]; (b) inkjet-printed Ag S/D electrodes based on Al:Nd/Al₂O₃:Nd/a-IGZO with different droplet spaces: (A) 35 μm, (B) 40 μm, (C) 45 μm [294].

our research group has firstly proposed a novel active lubrication generator driven by the PMJD for space mechanisms in 2011, the feasibility of the active lubrication was proved by Liu et al. [150]. A novel bearing lubrication device embedded in the gyroscope's bearing system was designed, as shown in Fig. 18(a), which can make effective use of the limited space without increasing the system mass [67]. To jet the lubrication oil with high viscosity, a new PMJD based on flexible hinge was proposed [75]. Besides, a PMJD was designed to realize the trace redundant lubrication, as seen in Fig. 18(b). The volume of the oil droplet could reach 0.0147 μL with the droplet velocity of 0.91 m/s [320].

7. Summary and perspective

It is well known that the PMJDs are becoming an increasingly additive manufacturing technology with the development of biomedicine and materials chemistry, microelectronic device, 3D printing, microelectronic packaging and space mechanism lubrication, due to their non-contact, maskless, compact structure, low cost, rapid dynamic response, high precision, convenient and environmental friendly. The origin of the PMJDs is introduced and the data analyses are counted in this review. This work can provide a comprehensive overview of the state-of-the-art in the PMJDs over the past few decades. The fundamental actuation modes and typical structure configurations are summarized. Simultaneously, the analysis methods of the PMJDs are also introduced, which can decrease the cost greatly and avoid unnecessary energy waste effectively.

The final goal of the PMJDs is to form the droplet with a desired volume and velocity, higher productivity, higher uniformity, more kinds of objects. Although the PMJDs have become mature gradu-

ally, there are still many challenges to be solved to achieve the ideal droplet characteristics. Firstly, the high droplet uniformity is difficult to be guaranteed because of air bubble entrapment, cross-talk effect and changing dynamics. Secondly, the droplet productivity is seriously limited due to residual vibration, channel fluid reflux and satellite droplet. Finally, the high temperature and high viscosity fluid are difficult to be jetted because of a small displacement deformation and a low Curie temperature of piezoelectric element.

Aiming at the current challenges, the schemes on improving the droplet uniformity and productivity are carried out. In structural optimization aspect, the diameter and layout of the nozzle are optimized to obtain an ideal droplet size, and the channel structure is also optimized to restrain the channel fluid reflux. However, the structural optimization increases processing difficulty and cost greatly. Alternatively, some pulse signals including sinusoidal, triangular, trapezoidal and square waveform are designed and optimized to generate the ideal droplets. But the droplet productivity is severely reduced due to the residual vibration. To suppress the residual vibration, some novel pulse signals such as bipolar, M-shaped and W-shaped waveform are discussed, which are achieved by adding an additional pulse waveform on the basis of the trapezoidal pulse waveform. In addition, the control strategies such as PID, feed-forward, iterative learning control and adaptive wavelet neural network are summarized to overcome the residual vibration and minimize the channel cross-talk effect.

Air bubble entrapment is also known to be major cause of the malfunction due to the poor droplet uniformity and productivity, and the malfunctions also include fluid degradation, aging, environmental temperature and nozzle blockage. Any problems in the jetting process must be identified and solved immediately. Based on this, fault detecting methods including vision-based method, self-

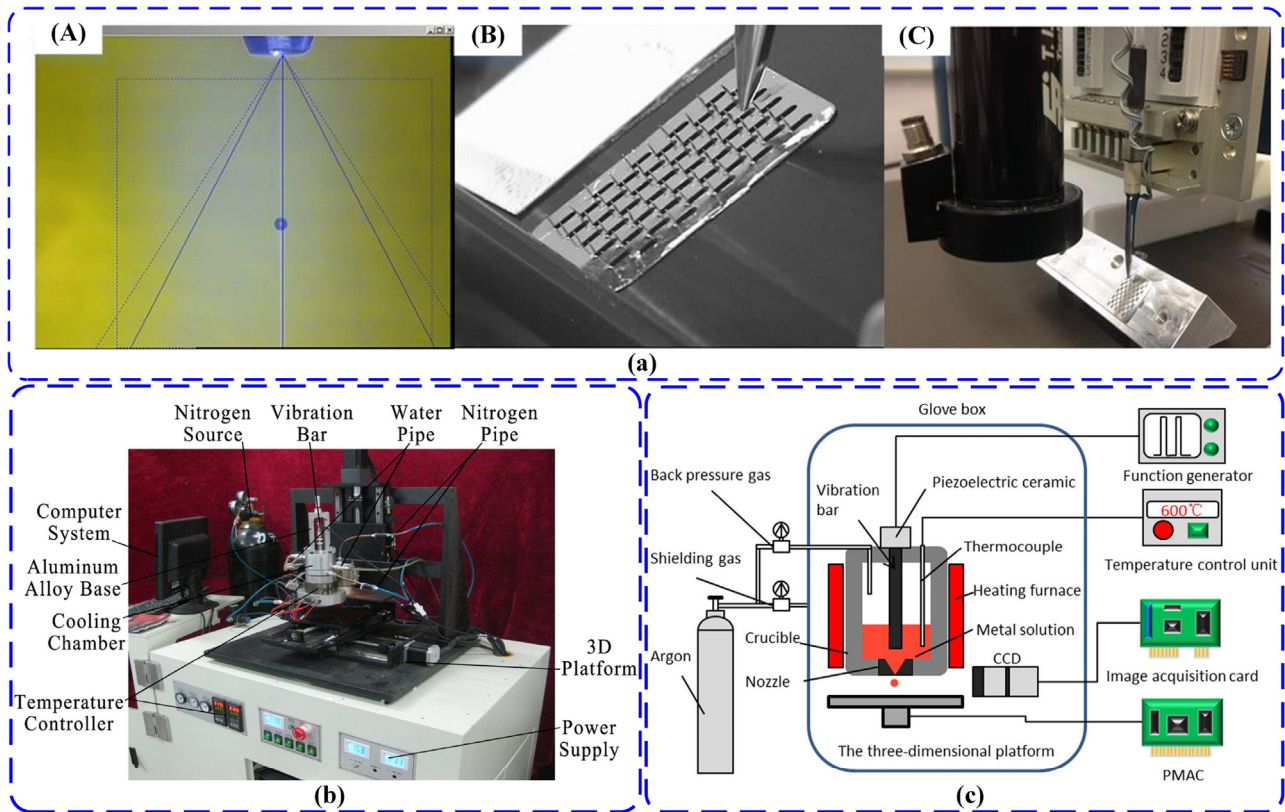


Fig. 17. 3D printing technology applications of the PMJDs: (a) The printing process of the anticancer agents: (A) Stroboscope image capture for measuring the droplet diameter, (B) the PMJD printing coating formulation on the microneedle array, (C) the printing configuration for the coating microneedles. The array is placed on a holder at an angle of 45° [303]; (b) the 3D printing system for the molten lead-free solder [313]; (c) the piezoelectric-actuated 3D printing system for a metal solution [315].

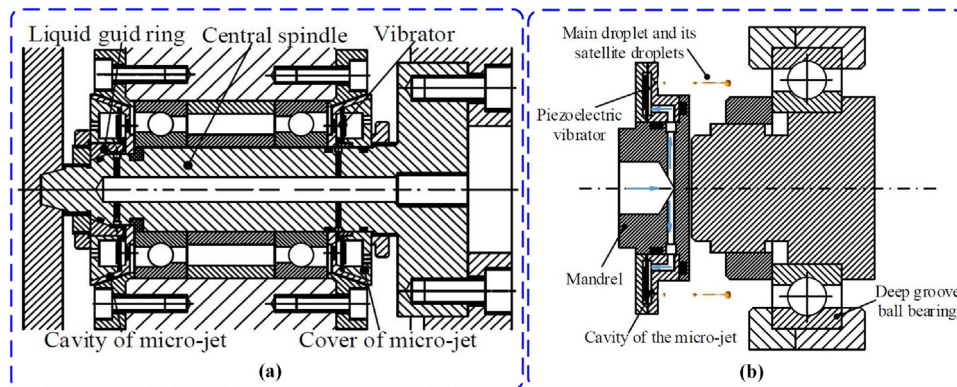


Fig. 18. Space mechanism lubrication of the PMJDs: (a) Structure of the gyroscope after embedding PMJD [67]; (b) Structure and lubrication principle of aerospace bearings using PMJD [320].

sensing technology and meniscus motion monitor are performed to monitor the droplet conditions. Finally, fluid characteristic analysis are discussed to get a clear comprehension on the relationship between fluid physical properties and droplet dynamic behaviors. The fluids used as the functional materials must be printable and stable. However, there are still many researches to be completed and optimized to solve the bottlenecks faced by the PMJDs.

Moreover, the future developments of the PMJDs are suggested as follows,

(1) PMJDs with multi piezoelectric actuators to achieve active control of acoustic pressure wave. The PMJDs are mainly driven by a single actuator to jet a wanted droplet. However, the residual vibration, channel fluid reflux and air bubble are easily produced. According to the acoustic mechanism, the acoustic pressure wave

can be well adjusted by the coordination control of the multi piezoelectric actuators. It is predicted that the researches on the PMJDs based on multi piezoelectric actuators will be carried out.

(2) The jetting behavior of high temperature and high viscosity fluid. With the rapid advance of the 3D metal printing, the study of high temperature fluid will enter a golden age. However, high temperature will make piezoelectric element work invalid because of its low Curie temperature. For high viscosity fluid, the shear stress between the inner wall of the nozzle and the fluid must be overcome to achieve a normal droplet process. However, the shear stress is too large to achieve the normal droplet operation because of a small displacement deformation of piezoelectric element. Although some needle-collision type PMJDs have been designed, the droplet frequency is limited and the configuration

is difficult to be miniaturized. The current bottlenecks will greatly stimulate the enthusiasm of the researchers in the high viscosity and the high temperature fluid.

(3) The driving method for the PMJDs. The PMJDs are generally driven by a trapezoidal waveform. However, the residual vibration of the pressure chamber is generated. To suppress the residual vibration, some novel pulse waveforms have been designed. Thus, it is expected that more and more researches on the driving method will be developed.

(4) PMJDs based on the special flow field such as venturi tube. The current PMJDs are designed and fabricated based on conventional flow channel structure and the main researches focus on the size parameters optimization of channel structure. It is imagined that the structure of the PMJDs is designed into a specific flow channel, and the special flow field will be produced. A qualitative leap in output performances may be realized. This method will break through the traditional ideas of structural design and optimization.

(5) The control strategy for the PMJDs. The current control strategies include feed-forward, PID, iterative learning control and adaptive wavelet neural network. With the rapid advance of intelligent algorithm, the new control strategy will be proposed.

(6) Research on the applications of the PMJDs such as biomedicine, 3D printing and space mechanism lubrication. In biomedical aspect, it is still in its infancy. Finding a safe and reliable PMJD for human body still poses a significant challenge, which will motivate the researchers to continue investigating a new PMJD and find a proper approach to incorporate them into biomedical systems. In 3D printing aspect, the piezoelectric 3D printing mainly includes fused deposition and selective laser melting. However, the fused deposition is low surface accuracy and it is difficult to print the metal sample. The selective laser melting is long processing cycle, complex process of the raw material recovery and utilization, high cost. The PMJDs are becoming more and more prominent in the 3D printing field as an efficient, economical and reliable technology. In space mechanism lubrication aspect, limited by high vacuum, high-low temperature and radiation, the lubrication system becomes the main factor in determining the reliability of space mechanism. The PMJDs will have wide research prospects in space mechanism lubrications, which can realize the active lubrication.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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